Journal of Combinatorial Optimization (2019) 37:424–438
https://doi.org/10.1007/s10878-018-0336-0

An improved lower bound on the growth constant of polyiamonds

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Published online: 16 August 2018
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Abstract
A polyiamond is an edge-connected set of cells on the triangular lattice in the plane. In this paper, we provide an improved lower bound on the asymptotic growth constant of polyiamonds, proving that it is at least 2.8424. The proof of the new bound is based on a concatenation argument and on elementary calculus. We also suggest a nontrivial extension of this method for improving the bound further. However, the proposed extension is based on an unproven (yet very reasonable) assumption.

Keywords Polyiamonds · Lattice animals · Growth constant

1 Introduction

A *polyomino* of size n is an edge-connected set of n cells on the square lattice $\mathbb{Z}^2$. Similarly, a *polyiamond* of size n is an edge-connected set of n cells on the two-dimensional triangular lattice. *Fixed* polyiamonds are considered distinct if they have different *shapes* or *orientations*. In this paper, we consider only fixed polyiamonds, and so we refer to them in the sequel simply as "polyiamonds". Figure 1 shows polyiamonds of size up to 5.

Work on this paper by all authors has been supported in part by ISF Grant 575/15. A preliminary version of this paper appeared in COCOON'17 (Barequet et al. 2017).

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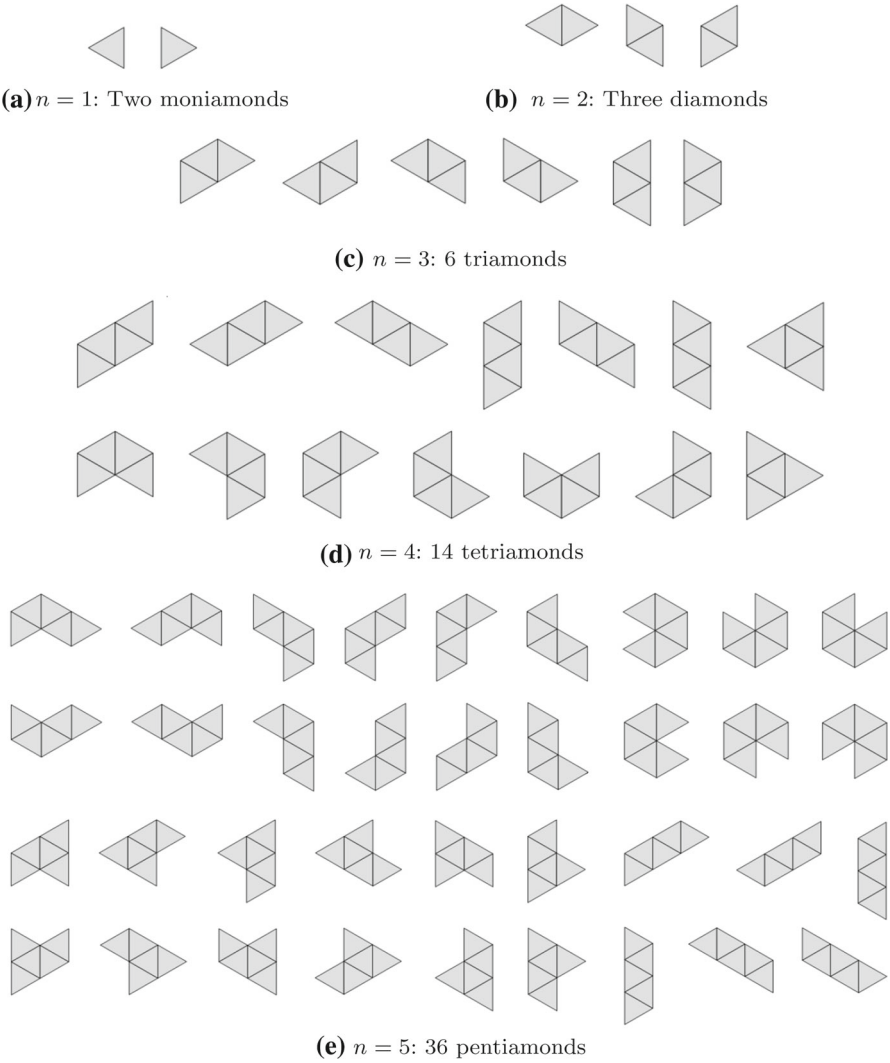

Fig. 1 Polyiamonds of sizes $1 \leq n \leq 5$

In general, a connected set of cells on a lattice is called a *lattice animal*. The fundamental combinatorial problem concerning lattice animals is "How many animals with n cells are there?" The study of lattice animals began in parallel more than half a century ago in two different communities. In statistical physics, (Temperley 1956) investigated the mechanics of macro-molecules, and (Broadbent and Hammersley 1957) studied percolation processes. In mathematics, (Harary 1960) composed a list of unsolved problems in the enumeration of graphs, and (Eden 1961) analyzed cell growth problems. Since then, counting animals has attracted much attention in the literature. However, despite serious efforts over the last 50 years, the problem of finding a formula for the number of polyominoes of a given size is still far from being solved,

and is considered (http://cs.smith.edu/~orourke/TOPP/) one of the long-standing open problems in combinatorial geometry.

The symbol $A(n)$ usually denotes in the literature the number of polyominoes of size n; See sequence A001168 in the On-line Encyclopedia of Integer Sequences (OEIS) (http://oeis.org). Since no analytic formula for the number of animals is yet known for any nontrivial lattice, a great portion of the research has so far focused on efficient algorithms for *counting* animals on lattices, primarily on the square lattice. Elements of the sequence $A(n)$ are currently known up to $n = 56$ (Jensen 2003). The growth constant of polyominoes was also treated extensively in the literature, and a few asymptotic results are known. Klarner (1967) showed that the limit $\lambda := \lim_{n\to\infty} \sqrt[n]{A(n)}$ exists, and the main problem so far has been to evaluate this constant. The convergence of $A(n+1)/A(n)$ to λ (as $n\to\infty$) was proven only three decades later by Madras (1999), using a novel pattern-frequency argument. The best-known lower and upper bounds on λ are 4.0025 (Barequet et al. 2016) and 4.6496 (Klarner and Rivest 1973), respectively. It is widely believed (see, e.g., Gaunt 1980; Gaunt et al. 1976) that $\lambda \approx 4.06$, and the currently best estimate, $\lambda = 4.0625696 \pm 0.0000005$, is due to Jensen (2003).

In the same manner, let $T(n)$ denote the number of polyiamonds of size n (sequence A001420 in the OEIS). Elements of the sequence $(T(n))$ are known up to $n = 75$ (Guttmann 2009, p. 479) using a transfer-matrix algorithm by Jensen [ibid., p. 173], adapting his original polyomino-counting algorithm (Jensen 2003). Earlier counts were given by Lunnon (1972) up to size 16, by Sykes and Glen (1976) up to size 22,[1] and by Aleksandrowicz and Barequet (2009) [extending Redelmeier's polyomino-counting algorithm (Redelmeier 1981)] up to size 31.

Similarly to polyominoes, the limits $\lim_{n\to\infty} \sqrt[n]{T(n)}$ and $\lim_{n\to\infty} T(n+1)/T(n)$ exist and are equal. Let, then, $\lambda_T = \lim_{n\to\infty} \sqrt[n]{T(n)}$ denote the growth constant of polyiamonds. Based on existing data, it is believed (Sykes and Glen 1976) (but has never been proven) that $\lambda_T = 3.04 \pm 0.02$.

Klarner (1967, p. 857) showed that $\lambda_T \geq 2.13$ by taking the square root of 4.54, a lower bound he computed for the growth constant of animals on the rhomboidal lattice, using the fact that a rhombus is made of two neighboring equilateral triangles. This bound is also mentioned by Lunnon (1972, p. 98). Rands and Welsh (1981/1982) used renewal sequences in order to show that

$$\lambda_T \geq (T(n)/(2(1+\lambda_T)))^{1/n} \tag{1}$$

for any $n \in \mathbb{N}$. Substituting the easy upper bound[2] $\lambda_T \leq 4$ in the right-hand side of Relation (1), and knowing at that time elements of $(T(n))$ for $1 \leq n \leq 20$ only [data

[1] Note that in this reference, the lattice is called "honeycomb" (hexagonal), and that the terms provided there should be doubled. The reason for this is that the authors actually count clusters of *vertices* on the hexagonal lattice, whose connectivity is the same as that of *cells* on the triangular lattice, with no distinction between the two possible orientations of the triangular cells. This is why polyiamonds are often regarded in the literature as site animals on the hexagonal lattice, and polyhexes (cell animals on the hexagonal lattice) are regarded as site animals on the triangular lattice, which sometimes causes confusion.

[2] This easy upper bound, based on an idea of Eden (1961) was described by Lunnon (1972, p. 98), and it is justified as follows. Every polyiamond P can be built according to a set of $n-1$ "instructions" taken from a superset of size $2(n-1)$. Each instruction tells us how to choose a lattice cell c, neighboring a cell

provided by Sykes and Glen (1976)], Rands and Welsh used $T(20) = 173,338,962$ to show that $\lambda_T \geq (T(20)/10)^{1/20} \approx 2.3011$. In fact, we wonder why Rands and Welsh did not use $T(22) = 1,456,891,547$ [which was also available to them (Sykes and Glen 1976)] to show that $\lambda_T \geq (T(22)/10)^{1/22} \approx 2.3500$. Nowadays, since we know $T(n)$ up to $n = 75$,[3] we can obtain, using the same method, that $\lambda_T \geq (T(75)/10)^{1/75} \approx 2.7714$. We can even do slightly better than that. Substituting in Relation (1) the upper bound $\lambda_T \leq 3.6108$ which we obtained elsewhere (Barequet et al. unpublished), we see that $\lambda_T \geq (T(75)/(2(1+3.6108)))^{1/75} \approx 2.7744$. However, we can still improve on this.

In this paper, we raise further the lower bound on λ_T, showing that $\lambda_T \geq 2.8424$. The new lower bound is obtained by combining a concatenation argument with elementary calculus. We also show how a reasonable assumption (however, without proof) can improve the lower bound even further.

2 Preliminaries

We orient the triangular lattice as is shown in Fig. 2a, and define a lexicographic order on the cells of the lattice as follows: A triangle t_1 is *smaller* than triangle $t_2 \neq t_1$ if the lattice column containing t_1 is to the *left* of the column containing t_2, or if t_1, t_2 are in the same column and t_1 lies *below* t_2. Triangles that look like a "left arrow" (Fig. 2b) are of Type 1, and triangles that look like a "right arrow" (Fig. 2c) are of Type 2. In addition, we say that a polyiamond is of Type i (for $i = 1, 2$) if its lexicographically-*largest* (top-right) triangle is of Type i, and denote by $T_i(n)$ the respective count of polyiamonds of size n.

Let $x(n)$ $(0 < x(n) < 1)$ denote the fraction of polyiamonds of Type 1 out of all polyiamonds of size n, that is, $T_1(n) = x(n)T(n)$ and $T_2(n) = (1 - x(n))T(n)$. In addition, let $y(n) = T_2(n)/T_1(n) = (1 - x(n))/x(n)$ denote the ratio, among polyiamonds of size n, between polyiamonds of Type 2 and polyiamonds of Type 1.

A concatenation of two polyiamonds P_1, P_2 is the union of P_1 and a translated copy of P_2, such that the lexicographically-largest triangle of P_1 is attached to the lexicographically-smallest triangle of P_2, and all cells of P_1 remain smaller than the translates of all cells of P_2. We use a concatenation argument in order to improve the lower bound on λ_T.

3 The bound

Our proof of a new lower bound on λ_T uses the division of polyiamonds into Type 1 and Type 2, but does not employ the asymptotic proportion between the two types.

Theorem 1 $\lambda_T \geq 2.8424$.

Footnote 2 continued
already in P, and add c to P. (Some of these instruction sets are illegal, and some other sets produce the same polyiamonds, but this only helps.) Hence, $\lambda_T \leq \lim_{n\to\infty} \binom{2(n-1)}{n-1}^{1/n} = 4$.

[3] $T(75) = 15,936,363,137,225,733,301,433,441,827,683,823$.

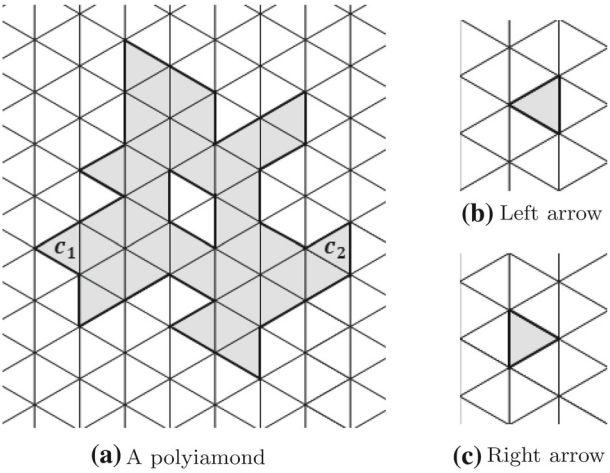

Fig. 2 Polyiamonds on the triangular lattice

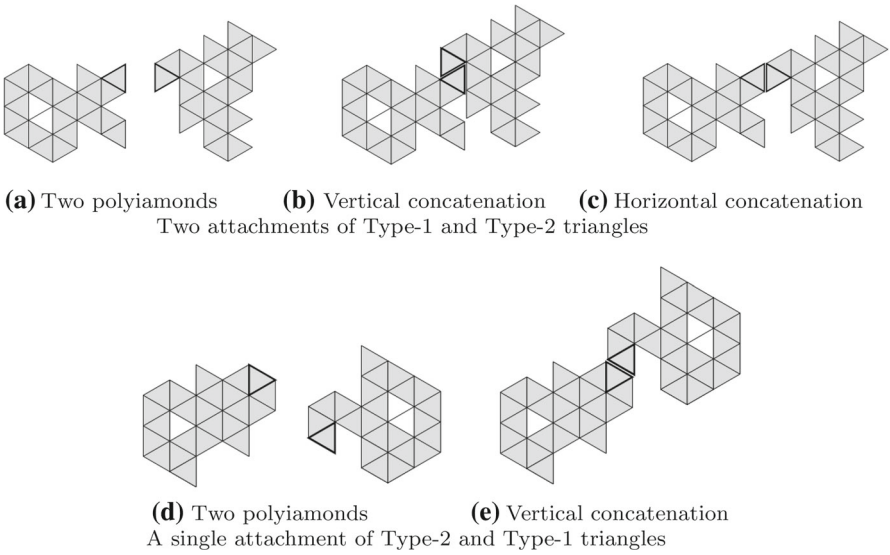

Fig. 3 Possible concatenations of polyiamonds

Proof First note that by rotational symmetry, the number of polyiamonds of size n, whose *smallest* (bottom-left) triangle is of Type 2, is $T_1(n)$. Similarly, the number of polyiamonds, whose *smallest* triangle is of Type 1, is $T_2(n)$.

We proceed with a concatenation argument tailored to the specific case of the triangular lattice. Interestingly, not all pairs of polyiamonds of size n can be concatenated. In addition, there exist many polyiamonds of size $2n$ which cannot be represented as a concatenation of two polyiamonds of size n. Let us count carefully pairs of polyiamonds that can be concatenated.

– Polyiamonds, whose largest triangle is of Type 1, can be concatenated only to polyiamonds whose smallest triangle is of Type 2, and this can be done in two different ways (see Fig. 3a–c). There are $2(T_1(n))^2$ concatenations of this kind.
– Polyiamonds, whose largest triangle is of Type 2, can be concatenated, in a single way, only to polyiamonds whose smallest triangle is of Type 1 (see Fig. 3d, e). There are $(T_2(n))^2$ concatenations of this kind.

Altogether, we have $2(T_1(n))^2 + (T_2(n))^2$ possible concatenations, and, as argued above,

$$2(T_1(n))^2 + (T_2(n))^2 \leq T(2n). \tag{2}$$

Let us now find an efficient lower bound on the number of concatenations. Relation (2) can be rewritten as $T(2n) \geq 2(x(n)T(n))^2 + ((1-x(n))T(n))^2 = (3x^2(n) - 2x(n) + 1)T^2(n)$. Elementary calculus shows that the function $f(x) = 3x^2 - 2x + 1$ assumes its minimum at $x = 1/3$, and that $f(1/3) = 2/3$. Hence,

$$\frac{2}{3}T^2(n) \leq T(2n).$$

By simple manipulations of this relation, we obtain that

$$\left(\frac{2}{3}T(n)\right)^{1/n} \leq \left(\frac{2}{3}T(2n)\right)^{1/(2n)}.$$

This implies that the sequence $\left(\frac{2}{3}T(k)\right)^{1/k}, \left(\frac{2}{3}T(2k)\right)^{1/(2k)}, \left(\frac{2}{3}T(4k)\right)^{1/(4k)}, \ldots$ is monotone increasing for any value of k, and, as a subsequence of $(\left(\frac{2}{3}T(n)\right)^{1/n})$, it converges to λ_T too. Therefore, any term of the form $\left(\frac{2}{3}T(n)\right)^{1/n}$ is a lower bound on λ_T. In particular, $\lambda_T \geq (\frac{2}{3}T(75))^{1/75} \approx 2.8424$. □

4 Convergence, interval containment, and monotonicity

First, we show a simple relation between the number of polyiamonds of Type 2 of size n and the number of polyiamonds of Type 1 of size $n+1$,

Observation 1 *$T_2(n) = T_1(n+1)$ (for all $n \in \mathbb{N}$).*

This simple observation follows from the fact that if the largest triangle t of a polyiamond P is of Type 1, then the only possible neighboring triangle of t within P is the triangle immediately below it, and so, removing t from P will not break P into two parts. This implies a bijection between Type-1 polyiamonds of size $n+1$ and Type-2 polyiamonds of size n.

Next, we prove the convergence of the sequences $(x(n))$ and $(y(n)) \equiv (T_1(n+1)/T_1(n))$, and find their limits.

Theorem 2 *The sequence $(T_1(n+1)/T_1(n))_{n=1}^{\infty}$ converges.* □

Indeed, polyiamonds of Type 1 fulfill all the premises of Madras's Ratio Limit Theorem (Madras 1999, Thm. 2.2). Hence, we can apply the mentioned theorem and conclude that $\lim_{n\to\infty} T_1(n+1)/T_1(n)$ exists and is equal to some unknown constant μ, whose value is specified indirectly in the next theorem. (See the "Appendix" for a detailed proof of Theorem 2).

Theorem 3 (i) $\mu = \lambda_T$;
(ii) $\lim_{n\to\infty} y(n) = \lambda_T$;
(iii) $\lim_{n\to\infty} x(n) = 1/(\lambda_T + 1)$.

Proof (i) On the one hand, by Madras (1999) we know that $\lambda_T := \lim_{n\to\infty} T(n+1)/T(n)$ exists. On the other hand, we have that

$$\frac{T(n+1)}{T(n)} = \frac{T_1(n+1)+T_2(n+1)}{T_1(n)+T_2(n)} = \frac{T_1(n+1)+T_1(n+2)}{T_1(n)+T_1(n+1)}$$
$$= \frac{1+\frac{T_1(n+2)}{T_1(n+1)}}{\frac{T_1(n)}{T_1(n+1)}+1} \xrightarrow[n\to\infty]{} \frac{1+\mu}{\frac{1}{\mu}+1} = \mu.$$

(Note that the convergence in the last step relies on the fact that the sequence $(T_1(n+1)/T_1(n))$ has a limit μ.) Hence, $\mu = \lambda_T$.

(ii)

$$y(n) = \frac{T_2(n)}{T_1(n)} = \frac{T_1(n+1)}{T_1(n)} \xrightarrow[n\to\infty]{} \mu = \lambda_T,$$

where we use Observation 1 and Theorems 2,3(i).

(iii)

$$x(n) = \frac{T_1(n)}{T(n)} = \frac{T_1(n)}{T_1(n)+y(n)T_1(n)} = \frac{1}{1+y(n)} \xrightarrow[n\to\infty]{} \frac{1}{\lambda_T+1},$$

where we use Theorem 3(ii). □

Now, we show relations between bounds on the entire sequence $(x(n))$ and bounds on the entire sequence $(T(n+1)/T(n))$. Note that the theorem does not argue that any such bound exists, but only that *if* one bound exists, *then* the other bound exists as well.

Theorem 4 [4] *Let d, e be two constants strictly greater than 1.*

(i) $x(n) \geq 1/d$ *for all* $n \in \mathbb{N} \longrightarrow T(n+1)/T(n) \leq d-1$ *for all* $n \in \mathbb{N}$;
(ii) $T(n+1)/T(n) \leq d-1$ *for all* $n \in \mathbb{N} \longrightarrow x(n) \geq 1/d$ *for all* $n \in \mathbb{N}$, *except, perhaps, and possibly infinitely many, isolated (no two consecutive) values of* n.
(iii) $x(n) \leq 1/e$ *for all* $n \in \mathbb{N} \longrightarrow T(n+1)/T(n) \geq e-1$ *for all* $n \in \mathbb{N}$.

[4] The preliminary version of this paper (Barequet et al. 2017) contained an error, claiming that the logical entailment in item (i) of this theorem is bidirectional.

(iv) $T(n+1)/T(n) \geq e-1$ *for all* $n \in \mathbb{N} \longrightarrow x(n) \leq 1/d$ *for all* $n \in \mathbb{N}$, *except, perhaps, and possibly infinitely many, isolated (no two consecutive) values of* n.

Note that in Theorem 4(i, iii), the premise of the logical consequence is *stronger* than the conclusion. This is because the condition about the ratio of successive elements of a sequence ($(T_1(n))$, in this case) is stronger than the same condition about the ratio of *sums of successive pairs* of elements of the same sequence. Therefore, in the opposite direction [Theorem 4(ii, iv)], we are able to prove only weaker versions of the premises in Theorem 4(i, iii).

Proof (i) First, we manipulate the premise of the logical consequence.

$$\begin{aligned} x(n) \geq \frac{1}{d} &\longleftrightarrow \frac{T_1(n)}{T(n)} \geq \frac{1}{d} \longleftrightarrow \frac{T_1(n)}{T_1(n)+T_1(n+1)} \\ &\geq \frac{1}{d} \longleftrightarrow \frac{T_1(n)+T_1(n+1)}{T_1(n)} \leq d \longleftrightarrow 1+\frac{T_1(n+1)}{T_1(n)} \\ &\leq d \longleftrightarrow \frac{T_1(n+1)}{T_1(n)} \leq d-1. \end{aligned}$$

Second, we prove the conclusion of the entailment.

$$\begin{aligned} \frac{T(n+1)}{T(n)} &= \frac{T_1(n+1)+T_1(n+2)}{T_1(n)+T_1(n+1)} = \frac{1+T_1(n+2)/T_1(n+1)}{T_1(n)/T_1(n+1)+1} \\ &\leq \frac{1+(d-1)}{1/(d-1)+1} = d-1. \end{aligned}$$

(ii) Suppose to the contrary that the claim is not true for two successive values of n. That is, there exists $n_0 \in \mathbb{N}$ for which $x(n_0) < 1/d$ *and* $x(n_0+1) < 1/d$. Using the same manipulation as in item (i), it is obvious that this is equivalent to assuming that $T_1(n_0+1)/T_1(n_0) > d-1$ and $T_1(n_0+2)/T_1(n_0+1) > d-1$. Now, an easy calculation shows that

$$\begin{aligned} \frac{T(n_0+1)}{T(n_0)} &= \frac{T_1(n_0+1)+T_1(n_0+2)}{T_1(n_0)+T_1(n_0+1)} = \frac{1+T_1(n_0+2)/T_1(n_0+1)}{T_1(n_0)/T_1(n_0+1)+1} \\ &> \frac{1+(d-1)}{1/(d-1)+1} = d-1, \end{aligned}$$

which contradicts the premise of the logical entailment.

(iii, iv) The proofs are completely analogous to those of items (i,ii), respectively. □

Finally, we observe the relation between the directions of monotonicity (if they exist) of the sequences $(x(n))$ and $(y(n))$.

Observation 2 *If either of the two sequences* $(x(n))$ *and* $(y(n))$ *is monotone, then the other sequence is monotone as well but in the opposite direction.*

Indeed, this follows immediately from the equality $y(n) = (1 - x(n))/x(n) = 1/x(n) - 1$. The available data suggest that $(x(n))$ be monotone decreasing and that $(y(n))$ be monotone increasing. Note that the monotonicity of $x(n)$ (or $y(n)$) neither implies, nor is implied by, the monotonicity of $(T(n+1)/T(n))$.

5 A conditional lower bound

Let $L(n)$ denote the number of animals on a lattice $\mathcal{L}$. It is widely believed (Whittington and Soteros 1990) (but has never been proven) that asymptotically

$$L(n) \sim C_{\mathcal{L}} n^{-\theta_{\mathcal{L}}} \lambda_{\mathcal{L}}^{n}, \tag{3}$$

where $C_{\mathcal{L}}$, $\theta_{\mathcal{L}}$, and $\lambda_{\mathcal{L}}$ are constants which depend on $\mathcal{L}$.[5] If this is true, it will imply at once the existence of the limit $\lambda_{\mathcal{L}} := \lim_{n\to\infty} L(n+1)/L(n)$, the growth constant of animals on the lattice $\mathcal{L}$. As was mentioned in the introduction, the existence of this limit was proven by Madras (1999) without assuming the asymptotic relation in Eq. (3). Furthermore, it is widely believed that the "ratio sequence" $(L(n+1)/L(n))$ is *monotone increasing*.[6] Available data (numbers of animals on various lattices) support this belief, but it was unfortunately never proven. Such monotonicity of the ratio sequence would imply that the entire sequence lies below its limit. (In fact, this may be the case even without monotonicity.) Consequently, every element of the ratio sequence would be a lower bound on the growth constant of animals on $\mathcal{L}$. In particular, it would imply, for the triangular lattice, the lower bound $\lambda_T \geq T(75)/T(74) \approx 2.9959$, only 0.05 short of the estimated value.

Let us restrict our attention, again, to the triangular lattice. Another plausible direction for setting a lower bound on λ_T would be to prove that the entire ratio sequence of $T(n)$ lies below some constant $c_1 > \lambda_T$, i.e., that $T(n+1)/T(n) \leq c_1$ for all $n \in \mathbb{N}$. As we demonstrate later in this section, proving a relation of this kind is not sufficient, and we need to prove a slightly-stronger claim, specifically, that the entire ratio sequence of $x(n)$ lies above some positive constant c_2. Proving this for *any* arbitrary value of c_2, even a very small one, would improve the lower bound on λ_T which is provided in Sect. 3.

In the proof of Theorem 1, we considered the three possible kinds of concatenations of two polyiamonds of size n. In fact, more compositions, which preserve the lexicographical order of the polyiamonds, can be obtained by attaching horizontally triangles of Type 1 and Type 2, which are not the largest or smallest triangles of the respective polyiamonds. The following cases, shown in Fig. 4, are categorized by the type of extreme triangles (t_1, t_2) in a pair of composed polyiamonds, where t_1 is the largest triangle of the smaller polyiamond, and t_2 is the smallest triangle of the larger

[5] In fact, it is widely believed (but has never been proven either) that the constant θ is common to *all* lattices in the same dimension. In particular, there is evidence that $\theta = 1$ for all lattices in two dimensions.

[6] Madras (1999, Prop. 4.2) proved "almost monotonicity" for all lattices, namely, that $L(n+2)/L(n) \geq (L(n+1)/L(n))^2 - \Gamma_{\mathcal{L}}/n$ for all sufficiently large values of n, where $\Gamma_{\mathcal{L}}$ is a constant which depends on $\mathcal{L}$. Note that proving that $\Gamma_{\mathcal{L}} = 0$ would imply that $L(n+2)/L(n+1) \geq L(n+1)/L(n)$, i.e., that the ratio sequence of $\mathcal{L}$ is monotone increasing.

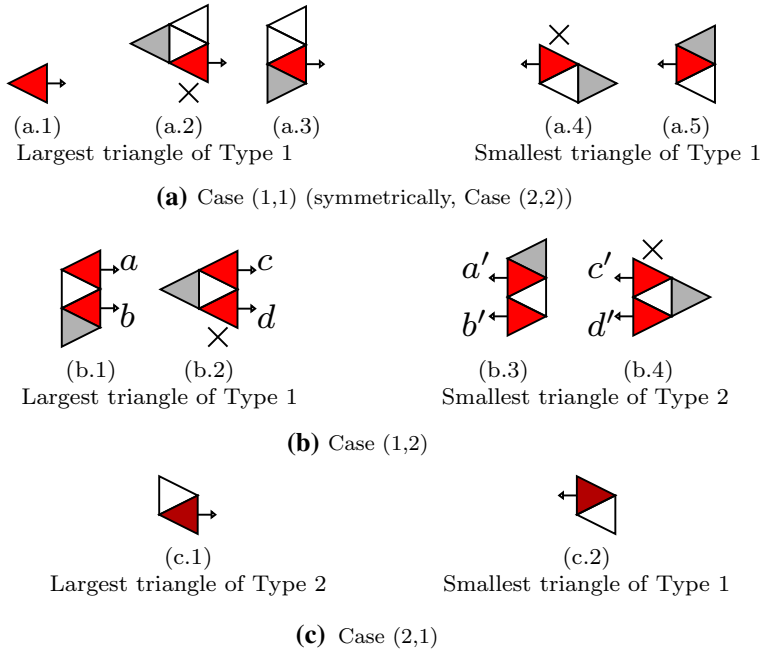

Fig. 4 Additional compositions of polyiamonds (Color figure online)

polyiamond. Therefore, these cases are distinct. In the figure, attached triangles are colored red, the direction of the attachment is marked by a small arrow, and an "×" sign marks an area free of triangles of the polyiamonds. The triangles in gray are the largest (or smallest) in polyiamonds of size less than n, which are then extended to full size-n polyiamonds. (In Case (c), the red and gray triangles identify.) For the purpose of further computation, let x_i denote the ratio $T_1(n-i)/T(n-i)$, for $i = 0, 1, 2, 3$ (that is, $x_i = x(n-i)$ with the omission of the reference to n), and let c be an upper bound on the entire sequence $(T(n+1)/T(n))$. All compositions below do not appear in the proof of Theorem 1.

(1,1) In this type of composition, each one of the polyiamond types shown in Fig. 4a.1–a.3 is composed with each one of the types shown in Fig. 4a.4, a.5. Specifically, the attached triangle in types (a.1) and (a.2,a.3) is the largest and third largest, respectively, and the attached triangle in types (a.4,a.5) is the second smallest. The number of polyiamonds whose largest triangle is of Type 1 (Fig. 4a.1) is $T_1(n)$, and among these polyiamonds, the number of polyiamonds shown in Fig. 4a.2 is $T_1(n-3)$. One way to see it is that, for every polyiamond of size $n-3$ whose largest triangle t (shown in gray) is of Type 1, a column-like polyiamond composed of three triangles whose largest is of Type 1, is attached to the right of t. With a similar reasoning, there are $T_2(n-3)$, $T_1(n-2)$, and $T_2(n-2)$ polyiamonds of the types shown in Fig. 4a.3–a.5, respectively. Hence, the number of compositions in this category is

Table 1 Counts of additional compositions in Case (1, 2)

Attached triangles	Number of compositions
(a ↔ a', c')	$T_2^2(n-3)+T_1(n-3)T_2(n-3)$
(b ↔ a', b', c', d')	$2T_2^2(n-3)+2T_1(n-3)T_2(n-3)$
(c ↔ a', c')	$T_1^2(n-3)+T_1(n-3)T_2(n-3)$
(d ↔ a', b', c', d')	$2T_1^2(n-3)+2T_1(n-3)T_2(n-3)$

$$\big(T_1(n)+T_1(n-3)+T_2(n-3)\big)\big(T_1(n-2)+T_2(n-2)\big).$$

By Observation 1, we can substitute $T_2(n-i)$ by $T_1(n-i+1)$, and by the definition of x_i, replace $T_1(n-i)$ by $x_iT(n-i)$. Then, by the definition of c, we have $T(n-i)\geq\frac{T(n)}{c^i}$, and conclude that in this category we have

$$\begin{aligned}&\big(x_0T(n)+x_3T(n-3)+x_2T(n-2)\big)\big(x_2T(n-2)+x_1T(n-1)\big)\\&\quad\geq\left(x_0+\frac{x_2}{c^2}+\frac{x_3}{c^3}\right)\left(\frac{x_1}{c}+\frac{x_2}{c^2}\right)T^2(n).\end{aligned}\tag{4}$$

Notice that, in principal, the attachment of smaller and larger triangles (of the two polyiamonds, respectively) can be counted as well. However, in such cases, higher powers of $1/c$ will appear in Eq. (4). The contribution of such compositions is relatively insignificant, thus, they are not considered.

(1,2) In this case, we count attachments of the largest and the third largest triangles in the smaller polyiamonds, and the smallest and third smallest triangles in the larger polyiamonds; see Fig. 4b.1–b.4. Table 1 lists all possible attachments and their corresponding numbers of compositions. The total number of compositions of this category is

$$3T_1^2(n-3)+6T_1(n-3)T_2(n-3)+3T_2^2(n-3)\geq 3\left(\frac{x_2}{c^2}+\frac{x_3}{c^3}\right)^2T^2(n).$$

(2,1) The numbers of polyiamonds shown in Fig. 4c.1, c.2 are both $T_1(n-1)$. Therefore, there are

$$T_1^2(n-1)\geq\frac{x_1^2}{c^2}T^2(n)$$

compositions in this category.

(2,2) By rotational symmetry, the number of compositions in this category is the same as that in Case (1,1).

Summing up the four cases and the three cases in the proof of Theorem 1, we obtain the relation

$$\left(1 - 2x_0 + 3{x_0}^2 + 2\left(x_0 + \frac{x_2}{c^2} + \frac{x_3}{c^3}\right)\left(\frac{x_1}{c} + \frac{x_2}{c^2}\right) + \frac{x_1^2}{c^2} + 3\left(\frac{x_2}{c^2} + \frac{x_3}{c^3}\right)^2\right) T^2(n) \leq T(2n). \tag{5}$$

In order to set a good (high) lower bound on λ_T, we need good lower bounds on x_0, x_1, x_2, x_3 and a good upper bound on c. Such bounds will imply a relation of the form $\beta \cdot T^2(n) \leq T(2n)$, where β is a constant, and will allow us to proceed in the same way as in the proof of Theorem 1. If we denote the 4-variable polynomial multiplying $T^2(n)$ by $f(x_0, x_1, x_2, x_3)$, we find that $f(\cdot, \cdot, \cdot, \cdot)$ is monotone decreasing when x_1, x_2, x_3 decrease, so its infimum is obtained when $x_1 = x_2 = x_3 = 0$, which is useless.

In fact, the available data suggest that the sequence $x(n)$ be monotone decreasing with a limit around 0.25. This is not a surprise since by Theorem 3(iii), we know that $x^* := \lim_{n\to\infty} x(n) = 1/(\lambda_T + 1)$, and the (unproven) estimate $\lambda_T \approx 3.04$ implies that $x^* \approx 0.2475$. Assume, then, *without proof* that $x(n) \geq 0.2$ for all $n \in \mathbb{N}$. (This is the only **unproven** assumption that we make!) By Theorem 4(i), this implies that $T(n+1)/T(n) \leq 4$ for all $n \in \mathbb{N}$. (Note, however, that we could not assume the latter in order to conclude the former, as the opposite entailment, provided in Theorem 4(ii), is weaker.) Plugging these bounds on x_1, x_2, x_3 and c into Relation 5, we observe in the left-hand side the term $T^2(n)$ multiplied by the quadratic function $g(x_0) = 3x_0^2 - 1.875x_0 + 1.00519$. (All actual calculations were performed with a much higher precision.) Elementary calculus shows that $g(x_0)$ assumes its minimum at $x_0 = 0.3125$, and that $g(0.3125) = 0.7122$. Hence,

$$0.7122 \cdot T^2(n) \leq T(2n).$$

With manipulations similar to those in the proof of Theorem 1, it can be shown that the sequence $\left(\left(0.7122 \cdot T(2^i k)\right)^{1/(2^i k)}\right)_{i=0}^{\infty}$ is monotone increasing for any value of k, and, as a subsequence of $\left(\left(0.7122 \cdot T(n)\right)^{1/n}\right)$, it converges to λ_T as well. Therefore, any term of the form $\left(0.7122 \cdot T(n)\right)^{1/n}$ is a lower bound on λ_T. In particular, $\lambda_T \geq (0.7122 \cdot T(75))^{1/75} \approx 2.8449$, which is a slight improvement over the lower bound on λ_T, shown in Sect. 3, and is pending only the correctness of the assumption that $x(n) \geq 0.2$ for all $n \in \mathbb{N}$.

Acknowledgements We wish to thank Neal Madras for helpful discussions about the use of his pattern and ratio-limit theorems in the context of polyiamonds. We also thank an anonymous referee for pointing out a flaw in Theorem 4.

Appendix: convergence of the sequence $(T_1(n+1)/T_1(n))$

In 1999, Madras (1999) proved a *Ratio Limit Theorem*, showing the existence of the growth constant (limit of the ratio sequence) for animals on any periodic lattice $\mathcal{L}$, namely, that $\lim_{n\to\infty} L(n+1)/L(n)$ (where $L(n)$ is the number of animals on $\mathcal{L}$)

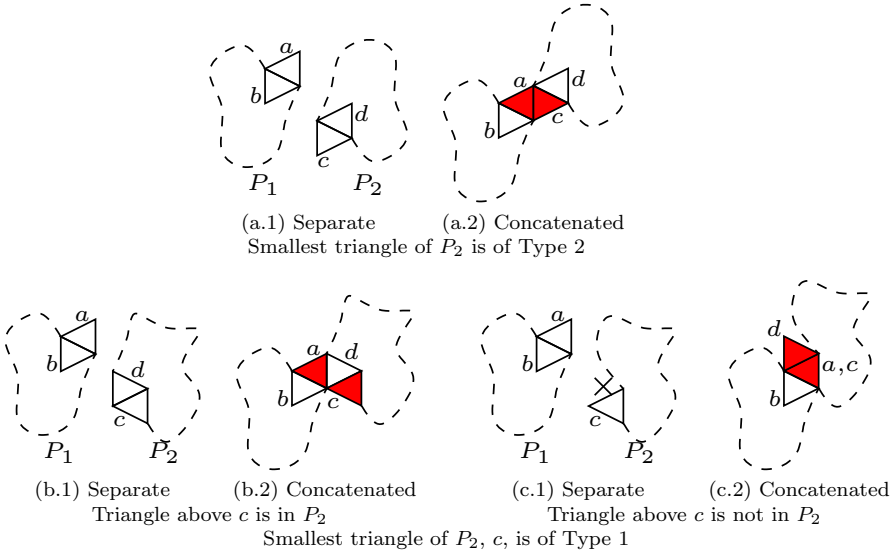

Fig. 5 Modified concatenation rules for Cluster Axiom 3 (Madras 1999) (Color figure online)

exists. The same method can be used to prove the existence of the growth constant of the sequences enumerating some restricted family of animals, provided that they fulfill some set of conditions. We show here the application of this method to polyiamonds of Type 1 on the planar triangular lattice.

Theorem 2 *The sequence* $(T_1(n+1)/T_1(n))_{n=1}^{\infty}$ *converges.*

Proof Our goal is to show that the set of polyiamonds of Type 1 fulfills all five Cluster Axioms and one additional condition, listed in the reference cited above.

The first axiom is held by definition: The set of polyiamonds of Type 1 of size n is a finite collection of subgraphs of the triangular lattice, which are invariant under translation.

The second axiom is also held in a trivial manner: If we assign the same weight to all polyiamonds of Type 1, then the difference in weight between "similar" such polyiamonds is obviously bounded.

The third axiom (the existence of $\lim_{n\to\infty}(T_1(n))^{1/n}$) is easily shown by a standard concatenation argument. The fact that the smallest triangle of some polyiamonds is of the "wrong" type, i.e., Type 1, can easily be overcome by modifying the concatenation rules. Refer to Fig. 5 which shows a few cases of concatenating two n-cell polyiamonds ($n > 1$) of Type 1. Let P_1 denote the left (lexicographically smaller) polyiamond. Since P_1 is of Type 1, its largest triangle, denoted by a, is of Type 1 by definition. The polyiamond P_1 must also contain a triangle b, of Type 2, located immediately below a. Let P_2 denote the right (lexicographically larger) polyiamond. If c, the smallest triangle of P_2, is of Type 2 (see Fig. 5a.1), then we attach c to the right side of a (Fig. 5a.2) as usual. Consider now the problematic cases, in which the smallest triangle of the right polyiamond P_2 is of Type 1. The polyiamond P_2 *may* contain a triangle d, of

Type 2, immediately above c. If triangle d is present in P_2 (see Fig. 5b.1), then we attach the left side of d to the right side of a (Fig. 5b.2). Although this operation is not a standard concatenation of two polyiamonds, it still yields a valid polyiamond of size $2n$. Otherwise, if triangle d is not present in P_2 (as in Fig. 5c.1), then we overlap triangles a and c, and add triangle d to the resulting shape (Fig. 5c.2), which is, again, a valid polyiamond of size $2n$. It is easy to verify that the resulting polyiamonds of size $2n$ in all cases are distinct. Indeed, in different cases, the positions of the $(n+1)$st triangle, relative to the nth triangle (both colored red in Fig. 5a.2, b.2, c.2), are different. The calculation in the concatenation argument now proceeds as usual.

Familiarity of the reader with the terminology and techniques used in Madras's work is required for the discussion of Axioms 4 and 5.

The fourth axiom is held as well. The axiom requires that any part of any cluster can be changed locally so as to create an occurrence of a translated copy of any so-called *proper pattern*.[7] Such a change is guaranteed to exist within the set of all polyiamonds, and, for our purposes, we need to show that it also exists within the set of polyiamonds of Type 1 only. Suppose, then, that we start from a polyiamond of Type 1 and the guaranteed change results in a polyiamond of Type 2. To fit Madras's terminology, we represent polyiamonds as connected subgraphs on the hexagonal lattice $\mathcal{H}$, which is the dual of the triangular lattice $\mathcal{T}$.[8] Madras describes how to change a part of any cluster (animal) to agree with a given pattern. His method uses a square bounding box, D, of the pattern, translated to the part of the cluster that is about to change. The inside of the box is replaced by the pattern, and the boundary of the box, ∂D, is added to the cluster in order to ensure the connectivity of the resulting animal. In our setting, the potential problem is that this operation may result in a polyiamond of Type 2, manifested on $\mathcal{H}$ by v, the largest vertex of the new animal, being of the wrong type. This may happen if v is not the largest vertex of the original animal (which is known to be of Type 1), but the largest vertex of ∂D. However, this can be avoided quite easily, either by choosing the origin of coordinates so that the top-right (largest) vertex of any axis-aligned square, whose corners have integer coordinates, is always of the desired Type 1, or by using, if necessary, a slightly-modified box, namely (on $\mathcal{H}$) the original box D plus one vertex located immediately above the top-right vertex of D.

The fifth axiom is also held as follows. This axiom requires that for any cluster, there always exist two proper patterns, whose unions of compulsory and forbidden parts are equal, such that if the cluster contains any of these patterns, then replacing in it this pattern by the other: (1) yields a valid (continuous) cluster; (2) reduces (resp., enlarges) the number of occurrences of the former (resp., latter) pattern in the cluster; (3) enlarges the size of the cluster by one. (The fourth assertion refers to the weight of clusters, but it is fulfilled trivially since in our context all clusters have the same

[7] A *proper pattern* is a subgraph P of the lattice, described by two parts: one which must be present in P, and the other, possibly empty, which must not be present in P.

[8] In this transformation, each triangle (cell) in $\mathcal{T}$ is mapped to a vertex (site) in $\mathcal{H}$, each edge (cell neighborhood) in $\mathcal{T}$ is mapped to an edge (bond) in $\mathcal{H}$, and each vertex in $\mathcal{T}$ is mapped to a hexagon (cell) in $\mathcal{H}$. Thus, a polyiamond on $\mathcal{T}$ is mapped bijectively to a cite animal on $\mathcal{H}$. The distinction between the two types of triangles on $\mathcal{T}$ is mapped to the distinction between the two types of vertices on $\mathcal{H}$, made by the relative orientation of the three bonds connected to a cite (namely, the orientation of the "fork"). The lexicographic order of triangles on $\mathcal{T}$ is mapped to the usual coordinates-based lexicographic order of vertices on $\mathcal{H}$.

weight.) Madras provides a pair of proper patterns which fulfills this axiom, again, with an argument involving a bounding box. As with the fourth axiom, we face the same problem that replacing the contents of a box in a polyiamond of Type 1 may result in a polyiamond of Type 2. However, this can be avoided in the same way described above.

Finally, we trivially have (using, again, a simple concatenation argument), that there exists a constant $c > 1$, for which we have that $T_1(n+1) \geq cT_1(n)$ for all $n \in \mathbb{N}$.

This is sufficient for showing that Madras's Ratio Limit Theorem (Madras 1999, Thm. 2.2, p. 367) can be applied to polyiamonds of Type 1. □

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